Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	*(a)*	(i)		Stefan-Boltzman law / Stefan's law (1) Substitution: $P = 4 \times \pi \times (7.0 \times 10^{8})^{2} \times 5.67 \times 10^{-8} \times (5\,800)^{4}$ (1) $P = 3.95 \times 10^{26}$ W seen (1)	1 1	1		3	2	
		(ii)	I	4	1			1		
			II	Positron [accept anti-electron]	1			1		
		(iii)		Production of 1 helium nucleus releases 26. 7 MeV = 4.27×10^{-12} J (1) $\frac{4.0 \times 10^{26}}{4.27 \times 10^{-12}} = 9.4 \times 10^{37}$ s^{-1} produced (1) [or 9.3×10^{37} s^{-1}] [Accept 9.2×10^{37} if 3.95×10^{26} used]		2		2	2	
	(b)			Substitution into $I = \frac{P}{4\pi R^2}$ i.e. $I = \frac{4.0 \times 10^{26}}{4\pi\left(1.5 \times 10^{11}\right)^2}$ (1) $I = 1\,415$ W (approx. 50% of surface intensity) (1) [1397 W using 3.95×10^{26} W]	1	1		2	2	
	(c)	(i)		Application of $P = IV$ to find max P from graph: Max $P = 6.4 \times 20.5 = 128$ W [accept 120 to 135 W] (1) % efficiency = $\frac{128 \times 100}{750}$ (1) % efficiency = 17% [Accept 16% to 18%] (1) [**Alt:** calculate 15% of 750 W and show is less]			3	3	2	
		(ii)		$\frac{1000}{128} = 7.8$ so 8 panels needed (1) [accept 8 or 9, depending on P output in (i)] [Answer must be whole number] **Mean** power will be less than 750 W (or equivalent) (1) Changing daily/seasonal conditions- need to be specific e.g. less light at night or less light in winter (1)		3		3		

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					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		E_k of particles is proportional to Temp (or $E_k = \frac{3}{2}kT$) (1) So particle E_k (or speed, velocity) needs to be high enough to overcome electrostatic (Coulomb) repulsion (1) ...and allow (nuclear) strong force to come into action (1)	1	1 1		3		
		(ii)		$n = \frac{2.2 \times 10^{22}}{75}$ [or $2.9(3) \times 10^{20}$ seen] (1) $2.9 \times 10^{20} \times 120 \times 10^{6} \times 0.8 = 2.8 \times 10^{28}$ Fusion possible (1)			2	2	2	
				Question 9 total	**6**	**9**	**5**	**20**	**10**	**0**